APPLIED AI CHALLENGE · PS-01 · TEAM REFERENCE

The PS-01 Bible

Everything about the AI-first POSP onboarding case, from A to Z, explained so simply that a five-year-old could follow it. It covers the case, the data, every number and where it came from, each slide, the jargon, and the questions judges will ask.

Case company: PolicyNest Insurance Brokers (fictional) · Problem owner: Vansh Miglani, InsuranceDekho
Built from: PS-01 brief, PS-01_posp_applications.xlsx, PS-01_document_review_test_set.xlsx, our insights workbook and the Check-in 1 deck · November 2026
All data in the pack is synthetic.

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0. How to use this book

Most sections use the same four kinds of boxes, so you always know what you are reading:

LIKE YOU'RE FIVE

The simplest possible explanation, with no jargon. If you can say this out loud, you understand it.

HOW WE GOT IT

The exact sheet, columns and formula. This is your proof when someone challenges a number.

IF A JUDGE ASKS

A likely question and a short, honest answer.

CAREFUL

A limit or caveat you must not overclaim.

PART 1

The whole story in 60 seconds

LIKE YOU'RE FIVE

Imagine a big **lemonade club**. Lots of people want to join and sell lemonade for the club. Before they can join, each person has to show **six papers**: a name card, an address card, a school report, a bank paper, a photo, and so on.

A grown-up checker looks at every paper **by eye**. But there are only a few checkers and lots of papers, so the papers sit in a **waiting pile for about 17 hours** before anyone even looks. The looking itself takes only **about 20 minutes**.

When the checker finds a problem, like a blurry photo, they send the person a message asking to fix **just that one thing**. The person fixes it and waits again. Then the checker finds the **next** problem. Each back-and-forth takes about **2 days**. Many people get tired and **give up**.

Out of every 100 people who start, only **58 end up selling lemonade**.

Our idea: a **robot helper** that looks at the papers the moment they arrive, spots every problem at once, asks for all the fixes in **one** message, and lets the grown-up checker handle only the truly tricky cases.

The same story, grown-up version

PolicyNest is an insurance broker. It sells insurance through about 14,000 part-time agents called **POSPs** (Point of Sales Persons). Before an agent can sell, a 22-person Onboarding Ops team checks their documents. Then the agent does 15 hours of training and passes an exam.

3,000
people started an application, Jan–Sep 2026

57.7%
got certified (1,732). 31.7% dropped out, 10.5% were rejected

8.4 days
median time from starting to certificate

17.3 h
median wait before a reviewer opens the file

Our one-line argument: the delay and the drop-off come from *how* documents are checked (late, by hand, one problem at a time), not from the checks themselves. An AI that checks at upload, asks once, and routes only judgement calls to people fixes the part of the journey we control.

Our four root causes:

1. **Late and one-at-a-time.** Problems are found after a 17-hour queue and asked for one by one. A returned file drops out 2.3× as often.
2. **Detectable but not detected.** 79% of the reasons files are sent back (blur, name, date of birth) could be caught when the photo is uploaded.
3. **Effort not aimed.** Reviewers spend the same ~21 minutes on clean and hard files, although 61% are clean approvals.

4. **Hand-off leak.** 482 people pass KYC and never start training. This is adjacent to our scope; we flag it to the owner.

Our success target: at least 60% of files decided correctly in minutes with no human touch, while catching at least 98% of the people who must be rejected.

PART 2

The case

2.1 Who is who

Name	What it is	Like you're five
InsuranceDekho	A real Indian insurance company. Vansh Miglani from there set the problem.	The person who gave us the homework.
PolicyNest Insurance Brokers	A *fictional* Gurugram broker made for this case. It sells motor, health and life insurance through ~14,000 POSPs in 14 states.	The pretend company in our story.
POSP	Point of Sales Person: a part-time agent (often a shopkeeper, mechanic or homemaker) who may sell simple insurance after 15 hours of training and an exam.	A helper who sells for the club.
Onboarding Ops team	22 reviewers in two teams (North; South & West) who check every new agent's documents.	The grown-up paper checkers.
Compliance officer	Makes sure no ineligible person is certified and every decision can be explained.	The rule-keeper who says "show me why".
IRDAI	The insurance regulator of India. Sets POSP rules.	The head teacher of all insurance clubs.

2.2 The four users we build for

User	What they need	How they'd use our product
Applicant (new POSP)	Get certified fast; know exactly what's wrong; fix it once.	Upload documents, get one clear message, re-upload, see status.
Reviewer	Stop re-reading clean files; spend time only on real judgement.	Work a queue of cases the AI couldn't decide, with conflicts highlighted.
Ops team lead	Meet service levels; see where files are stuck.	Watch queue age, AI decision mix, overrides and drop-off.
Compliance officer	No ineligible person certified; every decision explained.	Sample decisions; read the audit trail and the rule behind each one.

2.3 The six documents an applicant uploads

Document	What it proves	Like you're five

PAN card	Tax ID: name, date of birth, 10-character number. The 4th character shows the holder type (P = individual).	The main name card. Everything else is compared to it.
Aadhaar (front and back)	Identity and address. We must never store or show the full number.	The address card. Keep its number secret.
Class 10 marksheet	Education eligibility (Class 10 pass is the minimum).	The school report that proves you finished Class 10.
Cancelled cheque	Bank account for commission payouts. Must be in the applicant's own name.	Proof of the piggy bank where your earnings go.
Selfie	Face-match against the PAN photo (score 0–1).	A photo to check you are the person on the card.

2.4 The three possible decisions

Decision	Meaning	Test set share
APPROVE	Everything checks out; move to training.	364 / 600 = 60.7%
NEEDS_MORE_INFO	Fixable problem (blurry photo, DOB mismatch, cheque in a parent's name, spelling variant). Ask the applicant.	176 / 600 = 29.3%
REJECT	Not eligible: under 18, Class 10 not passed, PAN not an individual's, invalid PAN, already registered elsewhere, or a different person's document.	60 / 600 = 10.0%

LIKE YOU'RE FIVE

Three doors: "Yes, come in", "Almost, please bring one more thing", and "Sorry, you can't join". The golden rule: **if you're not sure, pick the middle door**. Wrongly saying "yes" breaks the law (a compliance breach). Wrongly saying "no" loses a good helper.

2.5 What we must build (from the brief)

Must have: read every document (with a confidence per field); cross-check name, DOB, PAN type, education, bank ownership and face-match; one application-level decision with every reason; one consolidated follow-up message in English or Hindi; send "Human only" decisions to a reviewer with evidence side by side; an evaluation on the 600-file test set (accuracy, plus precision and recall for REJECT and NEEDS_MORE_INFO); a working interface at a public URL.

Should have: re-check only the re-uploaded document; a reviewer dashboard; an audit trail; a simulation of the effect on time and drop-off.

Stretch: WhatsApp flow; DigiLocker; swap the rulebook to onboard a different customer type (e.g. loans).

Out of scope: building training or the exam; real e-KYC, penny-drop or IRDAI calls (use the pack's values); fraud investigation beyond flagging; storing full Aadhaar numbers.

2.6 The ten acceptance tests (T1–T10), in plain words

#	Situation	What our product must do
T1	Everything matches exactly	Approve in one pass, no human, and record every check
T2	Marksheet says "R K SHARMA", PAN says "RAHUL KUMAR SHARMA"	Treat initials as a match and approve; ask nothing
T3	MOHAMMED vs MOHAMMAD	NEEDS_MORE_INFO (NAME_SPELLING_VARIANT): ask for a corrected document or declaration
T4	Aadhaar and marksheet carry someone else's name	Recommend REJECT (NAME_MISMATCH), send to a human, never auto-reject
T5	Blurry marksheet and cheque in a parent's name	Ask for both fixes in **one** message
T6	PAN 4th character is C (a company)	REJECT (PAN_NOT_INDIVIDUAL) and explain in plain words
T7	Aadhaar DOB one year off PAN DOB	NEEDS_MORE_INFO (DOB_MISMATCH); never silently pick one
T8	Face-match score 0.68	Send to a human; don't decide alone
T9	Address state differs from declared state	Note it as informational; approve
T10	Corrected marksheet re-uploaded	Re-check only what changed; keep an audit trail of both rounds

2.7 The official success targets (from the brief)

Metric	Baseline today	Target
Decision accuracy	Reviewers 97.5% (document level)	≥ 95% vs ground truth
Recall on REJECT	Not measured	≥ 98%
Straight-through rate	0%	≥ 60%
Follow-up rounds per returned file	1–3, one at a time	1
Time to decision	17.3 h median queue + review	< 10 min for straight-through
Reviewer minutes per file	21	< 5 averaged over all files
Applicant clarity	Not measured	≥ 4/5 from five real users

2.8 Guardrails (the rules we must never break)

- Follow the **DPDP Act 2023**: collect only what's needed, state the purpose, and never send full Aadhaar numbers to a model, log or prompt.
- No real people's documents; use the synthetic data or our own specimens.
- AI recommends; a human confirms every NAME_MISMATCH rejection and every "Human only" decision.
- Every decision cites the rule it applied. "A decision you cannot explain is not a decision."
- Prefer NEEDS_MORE_INFO over REJECT when unsure.
- Check fairness across states, genders and name patterns, and report any gap.
- Applicant messages never mention fraud and never show another person's data.

2.9 Why now

LIKE YOU'RE FIVE

Robots have recently learned to **read photos of papers** and to **compare several papers at once**, for a few rupees per person. The government also built **DigiLocker**, a safe online cupboard where your papers already live, so people may not even need to take photos. And other clubs want the same helpers, so **whoever says "welcome" fastest keeps them**.

PART 3

The data pack

The brief lists four files. We had two workbooks plus the brief. The rulebook and the specimen documents were not part of our upload, so anything that depends on the rulebook's exact thresholds is marked as such.

3.1 PS-01_posp_applications.xlsx: today's process

LIKE YOU'RE FIVE

A diary of 3,000 people trying to join. One page has each person's details. Another page writes down **every single thing that happened** with a time stamp, like "10:02 uploaded papers" or "next day 09:15 checker opened file".

Sheet	Rows	What's in it
applications	3,018 (3,000 unique)	One row per application: channel, state, declared details, stage timestamps, status, rejection reason, reviewer, manual_touches, resubmissions, hours_to_certify
stage_events	21,188	Event log: application_id, event_time, actor (applicant / system / reviewer ID), event code, note
ops_team	22	Reviewers: team, tenure, shift, speed_index, loaded cost ₹/hour, applications handled
monthly_funnel	9	Monthly applications, certified, dropped, rejected, median hours, touches, certification rate

The 14 event codes, in journey order

Event	Count	What it means
APPLICATION_STARTED	3,000	Started on app (1,538), field referral (932) or web (530)
DOCS_SUBMITTED	2,742	All documents first uploaded
REVIEW_STARTED	2,742	Reviewer picked the file from the queue
DOC_RETURNED	849	Reviewer asked for a fix (774 unique applications)
DOC_REUPLOADED	638	Applicant sent a fix (582 unique applications)
REVIEW_RESUMED	638	Reviewer re-checked the fix
REJECTED	314	Not eligible (reason in the note)
KYC_VERIFIED	2,217	Documents verified and penny-drop passed

TRAINING_STARTED / COMPLETED	1,735	15-hour training
EXAM_FAILED	160	Scored below 40%; all of them later passed
EXAM_PASSED	1,735	Passed the POSP exam
CERTIFIED	1,732	Certificate issued; agent code active
AUTO_EXPIRED	951	System closed the file: no documents in 7 days (258), no reply to a fix request in 10 days (211), training not started in 14 days (482)

3.2 PS-01_document_review_test_set.xlsx: the answer key

LIKE YOU'RE FIVE

A practice exam with the **answers at the back**. It has 600 people's papers (3,600 papers in total), what a robot "read" from each paper, what the human checker said, and the **correct final answer**. We grade our robot against the correct answers, not against the human, because the humans are sometimes wrong.

Sheet	Rows	What's in it
documents	3,600	doc_type, image_quality (OK / BLURRY / CROPPED / GLARE), ocr_confidence, ocr_text, name/DOB/ID extracted, face_match_score (selfies), human_verdict, human_reason_code, reviewer_minutes
application_truth	600	Declared details, penny_drop_name_returned, irdai_posp_registry_hit, **final_decision** (ground truth), decision_reasons, name_match_category, is_hard_case
reason_codes	14	Codes like IMAGE_BLURRY → REUPLOAD, PAN_NOT_INDIVIDUAL → REJECT, FACE_MATCH_LOW (<0.60) → NEEDS_INFO

The seven name-match categories

Category	Example vs PAN "RAHUL KUMAR SHARMA"	Files	Correct = APPROVE
EXACT	RAHUL KUMAR SHARMA	436	285
INITIAL	R K SHARMA	28	18
MIDDLE_MISSING	RAHUL SHARMA	29	23
SWAP	SHARMA RAHUL KUMAR	36	23
MIDDLE_ADDED	RAHUL KUMAR PRASAD SHARMA	23	15
SPELLING	MOHAMMED vs MOHAMMAD	36	0 (34 need info)
DIFFERENT	A different person	12	0 (all 12 reject)

3.3 Deliberate traps in the data, and what we did

Trap	What we did
18 applications appear twice	Kept one row per application_id → 3,000
40 city names in UPPER CASE; 25 blank channels	Not used in headline numbers; blank channel shown as "Unknown"
Status is a snapshot on 30 Sep 2026	3 files still IN_PROGRESS; kept as their own bucket in the funnel
~2.5% of reviewer document verdicts are wrong	We always grade against application_truth.final_decision
No images; OCR text stands in and is corrupted when quality is poor	Noted as a risk: OCR confidence is 0.60 on poor images vs 0.93 on good ones
Light OCR noise (O read as 0, I as l) in ~25% of good documents	A reason to use an LLM that tolerates noise rather than exact string matching
Address state may differ from declared state	Informational only; never a reason to reject (test T9)

3.4 Our own insights workbook

PolicyNest_Onboarding_Data_Insights.xlsx has three sheets: **Executive Insights** (25 data points with meaning and product implication), **Problem to Solution** (5 problems mapped to solution, metric and caveat), and **Definitions & Caveats**. We re-computed every figure in it from the raw data and all of them matched.

PART 4

Today's journey, step by step

Times are medians between the first time each event happened for each person. Red steps are the "document-review zone", the part an AI can own.

1. Apply and upload
Types details, uploads 6 photos
17.9 h

2. Wait in queue
File sits in a shared pile
17.3 h

3. Reviewer checks
Compares names and dates by eye
~20 min

4. Fix asked
One problem per WhatsApp message
~2.2 days / loop

5. KYC verified
Penny-drop passes
23.3 h to training

6. Training
15 hours logged online
69.5 h

7. Exam and certificate
Pass exam, get code
19.8 h + 10.7 h

Step	What happens	Time (median)	Who we lose here
1. Apply and upload	Applicant types details and uploads PAN, Aadhaar, marksheet, cheque, selfie. Typing is never checked against documents.	17.9 h (P90 46.3 h)	258 never upload anything (8.6%); expired after 7 days
2. Queue	File waits in a shared queue.	17.3 h (P90 38.2 h, P95 47.1 h)	No one directly, but it delays everything
3. Review	Reviewer reads each document, compares names and dates, checks PAN format and face score.	~0.3 h elapsed; 20.7 min of work	314 rejected (10.5%), median 17.7 h after upload
4. Fix loop	Reviewer asks for one fix; applicant re-uploads; reviewer re-checks.	39.1 h to re-upload + 14.8 h to re-check ≈ 2.2 days	211 never reply (27.3% of 774 returned)
5. KYC → training	Penny-drop confirms the bank account; agent is sent to training.	23.3 h to start training	482 never start (21.7% of 2,217 verified)
6. Training	15 hours of online training.	69.5 h elapsed	No one (all who start, finish)
7. Exam → certificate	Exam (160 failed once, all later passed), then certificate.	19.8 h + 10.7 h	3 still in progress on 30 Sep

LIKE YOU'RE FIVE

The checking itself is quick: about the time it takes to eat lunch. Waiting for the checker takes as long as a whole night's sleep plus a whole school day. And every time the checker sends a paper back, add **two more days**.

Three facts that surprise people

1. **98% of the time before the first decision is waiting.** The queue takes 17.3 h and the review 0.3 h, so 17.3 / 17.6 ≈ 98%.

2. **Time of day doesn't matter.** The median wait is 16.6–18.0 h whatever hour a file arrives, and 15.8–18.9 h on every weekday. So this isn't a "nobody works at night" problem; it's a standing backlog.
3. **Training is the biggest single block of time** (37% of the average journey), and it is out of scope. So AI alone cannot get onboarding to 2 days. Our honest simulated target is ~6.9 days.

HOW WE GOT IT

stage_events → for each application, take the first timestamp of each event → subtract (later – earlier) in hours → median across applications. Appendix block [2].

PART 5

The number dictionary

Every number that appears in the deck: what it means simply, exactly how it was calculated, and where it's used. "Block" refers to the numbered blocks in the appendix script.

Number	Plain meaning	How calculated	Block · Slide
3,000	People who started an application	applications sheet, 3,018 rows, de-duplicated on application_id	0 · 3, 6
57.7%	Share who became agents	1,732 status = CERTIFIED ÷ 3,000	1 · 2, 3, 6
31.7% / 10.5%	Dropped / rejected	951 DROPPED ÷ 3,000; 314 REJECTED ÷ 3,000	1 · 6
1,268	People who never certified	3,000 − 1,732 (= 258 + 314 + 211 + 482 + 3)	1 · 1, 3
258	Never uploaded documents	AUTO_EXPIRED note "No documents uploaded within 7 days"	1 · 3, 4
211	Went silent after a fix request	AUTO_EXPIRED note "No response to document request in 10 days"	1 · 3, 6
482	Verified but never started training	AUTO_EXPIRED note "Training not started within 14 days"	1 · 3, 7
469 (37%)	Lost in the document stage	258 + 211 = 469; 469 ÷ 1,268 = 37.0%	1 · 1, 3
83%	Rejections that follow a fixed rule	EDU 87 + ALREADY_REG 75 + UNDERAGE 49 + PAN_INVALID 37 + PAN_NOT_INDIV 13 = 261 ÷ 314. Only NAME_MISMATCH_HARD (53) needs judgement	1 · 3, 11
17.7 h	How long a rejected applicant waits to hear "no"	Median DOCS_SUBMITTED → REJECTED	2 · 3, 14
53–63%	Monthly certification range	monthly_funnel.certification_rate, Jan–Sep (0.528 to 0.629)	4 · 3
8.4 days	Median time to certificate	Median APPLICATION_STARTED → CERTIFIED for 1,732 certified (8.41 d); matches hours_to_certify	2 · 2, 4, 6
6.6 / 13.0 d	Fastest quarter / slowest tenth	25th and 90th percentile of the same	2 · 5, 6
17.3 h	Wait before a reviewer looks	Median DOCS_SUBMITTED → REVIEW_STARTED (n = 2,742); P90 38.2 h, P95 47.1 h	2 · 2, 4, 6

0.3 h / ~20 min	The actual review	Median REVIEW_STARTED → DOC_RETURNED/KYC; test set reviewer_minutes 20.7 per application	2, 5 · 4, 5
98%	Share of pre-decision time spent waiting	17.3 ÷ (17.3 + 0.3)	2 · 5
15 h vs 76 h	Review zone, fast vs slow agents	Median DOCS_SUBMITTED → KYC_VERIFIED for fastest 25% vs slowest 10%	2 · 5
0.7 vs 3.3 d	Review zone, clean vs returned	Same interval, split by "had any DOC_RETURNED"	3 · 2, 4
~2.2 days	Cost of one fix loop	Median DOC_RETURNED → DOC_REUPLOADED (39.1 h) + DOC_REUPLOADED → REVIEW_RESUMED (14.8 h) = 53.9 h	2 · 4, 8
25.8%	Files sent back for a fix	774 unique applications with DOC_RETURNED ÷ 3,000	3 · 6
849	Return events	Count of DOC_RETURNED rows (703 apps once, 67 twice, 4 three times)	3 · 6, 9
71	Files returned more than once	67 + 4	3 · 7, 9
27.3%	Silence after a fix request	211 ÷ 774	1, 3 · 3, 6, 12
42.6% vs 18.4%	Drop-out, returned vs not	Among 2,742 who uploaded: DROPPED share for returned (330/774) vs not returned (363/1,968). 42.6 ÷ 18.4 = 2.3×	3 · 2, 8
10.8 vs 7.7 d	Days to certificate, returned vs not	Median hours_to_certify ÷ 24 by group; difference 3.1 days	3 · 8
4.7 vs 2.8	Human touches, returned vs not	Mean manual_touches by group (4.70 ÷ 2.82 = 1.7×)	3 · 8
58% vs 6.5%	Share of slow vs fast agents who had a return	Of the slowest 10% certified, 58.0% had a DOC_RETURNED; of the fastest 25%, 6.5%	3 · 4, 8
79% (668)	Returns a machine could flag at upload	Image 423 + name 130 + DOB 115 = 668 ÷ 849 = 78.7%	3 · 2, 7, 9
49.8%	Returns caused by image quality alone	423 ÷ 849	3 · 7, 14
181	Returns needing an outside service	Cheque name 111 + selfie 70 (penny-drop / face-match)	3 · 9
3.3	Human touches per file	Mean manual_touches; 9,920 in total	4 · 6

37%	Touches spent on returned files	3,640 ÷ 9,920; returned files are 26% of volume	3 · 6
~950 h / ₹4.1 L	Reviewer time on first review	2,742 files × 20.7 min ÷ 60 ≈ 946 h × ₹430/h ≈ ₹4.07 L. ₹430 = cost weighted by files handled per reviewer	4 · 6
60.7 / 29.3 / 10%	Correct decision mix in the test set	final_decision: 364 / 176 / 60 of 600	5 · 11, 12
20.6 vs 20.9	Reviewer minutes, clean vs hard	Sum of reviewer_minutes per application, mean by is_hard_case	5 · 7, 10
79 of 364 (22%)	Good approvals with a legitimate non-exact name	APPROVE where name_match_category is INITIAL 18 + MIDDLE_MISSING 23 + SWAP 23 + MIDDLE_ADDED 15	5 · 10, 13
80 / 53	"Reviewer discretion" codes, and how many were on approvable files	human_reason_code = REVIEWER_DISCRETION (80 documents); 53 of the applications they sit on are truly APPROVE	5 · 10, 14
39 of 176 (22%)	Fix-request cases with 2+ problems	NEEDS_MORE_INFO with 2+ semicolon-separated decision_reasons	5 · 9
22 of 60 / 9	Rejections a one-document-at-a-time check misses	Roll-up of reviewer verdicts vs truth: of 60 true REJECTs, 38 rolled up to REJECT, 13 to NEEDS_MORE_INFO and 9 to APPROVE. The 22 misses are age, registry or PAN-type rules	5 · 13, 14
0.60 vs 0.93	OCR confidence, poor vs good images	Mean ocr_confidence by image_quality	5 · 13
53% – 65%	Certification range across states	Odisha 53.2% … Delhi 64.5%	4 · 13, 14
8.4 → 6.9 d	Simulated median if the review zone took 10 min	For each certified journey: total − (DOCS_SUBMITTED → KYC) + 10 min; P90 13.0 → 10.8	6 · 12
~105 (+6%)	Extra agents if silence fell to 10%	Silent today 211; at 10% of 774 it would be 77 → 134 fewer. × 78.9% (share of repliers who certify) ≈ 106. 106 ÷ 1,732 ≈ 6%	6 · 12
≈ 13%	Return rate if image issues are prevented	48.7% of first returns are image quality → 774 × (1 − 0.487) ÷ 3,000 ≈ 13.2%	· 12

IF A JUDGE ASKS "WHY MEDIAN AND NOT AVERAGE?"

Waiting times have a long tail: a few people wait days, which drags the average up (queue: median 17.3 h, mean 20.4 h). The median is the "typical person". We also show P90 so the tail isn't hidden.

PART 6

Root causes

A root cause is the "why behind the why". Symptom: people quit. Cause: they were asked for fixes late, one by one. Root cause: nobody checks documents at the moment of upload.

RC1. Errors are found late and asked for one at a time

LIKE YOU'RE FIVE

It's like a teacher who waits until the next day to mark your homework, then tells you about **one** mistake. You fix it and wait another day, and she tells you about the next one. You'd give up too.

- Files wait 17.3 h before anyone looks (P90 38.2 h).
- 71 files were returned 2–3 times. The issue pairs (blurry + name: 19; DOB + blurry: 13; cheque + blurry: 10; blurry + selfie: 9; DOB + name: 5) were all visible at the first review.
- Each loop costs ~2.2 days.
- Returned files drop out at 42.6% vs 18.4%, take 10.8 vs 7.7 days, and need 4.7 vs 2.8 touches.
- 58% of the slowest 10% of agents had a return, against 6.5% of the fastest 25%.

What fixes it: check at upload; collect every issue; send one message.

RC2. Most problems are machine-detectable but aren't caught at upload

LIKE YOU'RE FIVE

Half of all "please fix" messages are "your photo is blurry". A phone can tell a photo is blurry **the second you take it**, like when your camera says "hold still".

- Return reasons: blurry or cropped 423 (49.8%), name spelling 130, DOB mismatch 115, cheque not in own name 111, selfie mismatch 70.
- Blur, name and DOB together make up 668 / 849 = 78.7%. These need only reading and comparing.
- Cheque name and selfie (181) need outside services (penny-drop, face-match), which our product would call.

CAREFUL

"Detectable" doesn't mean "auto-resolved". The applicant still has to fix it, but they find out in minutes, not days, and in one message.

RC3. Human effort is spread evenly, not aimed at judgement

LIKE YOU'RE FIVE

The checker spends the **same 21 minutes** on an easy paper as on a tricky one. Six out of ten papers are easy "yes" papers. That time could go to the tricky ones.

- Clean files 20.6 min, hard cases 20.9 min. True APPROVE 20.7, NEEDS_MORE_INFO 20.9, REJECT 20.2.
- 79 of 364 good approvals (22%) have a legitimate non-exact name (initials, missing or extra middle name, swapped order). An exact-match robot would wrongly stall 1 in 5 good agents.
- 80 documents were coded "REVIEWER_DISCRETION" with no rule cited; 53 of those were on files the rulebook would approve. This shows judgement drifting.
- About 2.5% of reviewer document verdicts are wrong (stated in the brief).

RC4. Hand-offs leak after verification (adjacent)

LIKE YOU'RE FIVE

Some people pass all the checks, get their "welcome" message, and then **never show up to class**.

- 482 of 2,217 verified (21.7%) never start training; median 23.3 h before those who do start.
- This is the single largest leak, but training is out of scope, so we raise it as an owner question.

Why these four (MECE check)

Together they cover **when** problems are found (RC1), **what** could be found automatically (RC2), **who** spends effort on what (RC3), and **what happens after** verification (RC4). They don't overlap, and between them they explain both the time and the drop-off.

PART 7

Scope, success, risks, owner questions

7.1 Scope: who decides what

AI decides, end to end	AI recommends, human confirms	Out of scope
Image-quality gate at upload	REJECT for a different person's name (never auto-reject)	Training and exam
Read name, DOB and ID from each document	Borderline face-match (e.g. 0.68)	Real e-KYC, penny-drop, IRDAI calls
Cross-check names (with rulebook variants) and DOB	Anything the rulebook marks "Human only"	Fraud investigation
PAN type, age ≥ 18, Class 10 pass	Low-confidence cases	Storing full Aadhaar
Use registry and penny-drop results		Adjacent v2: nudge the 482 who never start training
One consolidated fix message		

LIKE YOU'RE FIVE

The robot does the easy, rule-following jobs ("Is this person over 18? Is the photo clear?"). The grown-up does the "hmm, let me think" jobs ("Is this really the same person?").

Evidence: APPROVE + NEEDS_MORE_INFO are 90% of test-set files. Only 12 (2%) are different-person names. In production, 83% of rejections follow a fixed rule.

7.2 Success: the north star and the guardrail

LIKE YOU'RE FIVE

A **north star** is the one number you steer by. Ours: "How many files did the robot get right **all by itself**?" The **guardrail** is the fence you must not cross: "Never let in someone who isn't allowed."

- **North star, straight-through rate:** 0% → ≥ 60%. Realistic, because 60.7% of test files are true approvals.
- **Guardrail, REJECT recall:** ≥ 98% (catch at least 98 of every 100 people who must be rejected, or send them to a human).
- **Supporting metrics:** upload → decision < 10 min; one message per returned file; silence < 10%; returns ≈ 13%; reviewer minutes < 5; accuracy ≥ 95%.
- **Business outcome (simulation):** median 8.4 → 6.9 days; ~105 more agents per nine months (+6%).

7.3 Risks

Risk	Like you're five	Evidence	Mitigation

Wrongful approval (High)	Letting in someone who's too young	22 of 60 true rejections come from age, registry or PAN-type rules that no single paper shows. A paper-by-paper check would pass 9.	Hard-coded rule checks before any AI step; release only if REJECT recall ≥ 98%
Wrongful rejection (High)	Turning away "R K Sharma" because his card says "Rahul Kumar Sharma"	79 of 364 approvals have legitimate name variants; 36 more are spelling variants	Encode the rulebook's name rules; prefer NEEDS_MORE_INFO when unsure
Noisy photos (Med)	Robot can't read a blurry photo	OCR confidence 0.60 on poor images vs 0.93	Quality gate; per-field confidence; ask to retake
Unfairness (Med)	Robot works better for some states than others	Certification ranges 53% to 65% by state	Report accuracy by state, gender, name pattern
Privacy and trust (Med)	Keeping secret numbers secret; people may not trust a robot	DPDP Act; 27% already go silent	Mask Aadhaar; plain-language messages; never mention fraud

CAREFUL: THE "9 WOULD PASS" FINDING

This comes from our roll-up of reviewers' *document* verdicts. In real life, reviewers probably also check age and the registry separately. The point is not "reviewers miss these". The point is that **an engine that only reads documents one by one would miss them**, so application-level rules must be explicit.

7.4 Owner questions, and why each one matters

#	Question	Why we ask (the data)
1	What error rate on auto-approvals would compliance sign off, and who signs?	Sets how far straight-through can go. Reviewers are at 97.5% today.
2	Why does a 20-minute check wait 17 hours?	The wait is the same at every hour and on every weekday. Is it staffing, batching or SLA design?
3	Are registry and penny-drop checks run before or after review? What does each call cost?	22 of 60 rejections depend on them; rejections take 17.7 h.
4	What does "reviewer discretion" usually mean?	80 codes with no rule; 53 on approvable files.
5	Is the training hand-off in our remit?	482 (21.7%) never start: the biggest single leak.
6	Can the pilot use DigiLocker?	49.8% of returns are image quality.
7	Which pilot region, and should applicants be told an AI decided?	Certification varies 53–65% by state; 27% go silent.

Red questions must be answered before we lock targets, because they decide how much the engine may do on its own.

PART 8

Slide-by-slide guide

For each slide: what it shows, a script you can say in about a minute, and the question a judge is most likely to ask. Speak the title first. The title *is* the message.

1 Title



Say: "42 of every 100 people who apply to become a PolicyNest agent never get there. We'll show you that the document stage, not the work itself, is where they're lost, using 3,000 real-shaped applications and a 600-file answer key."

2 Executive summary

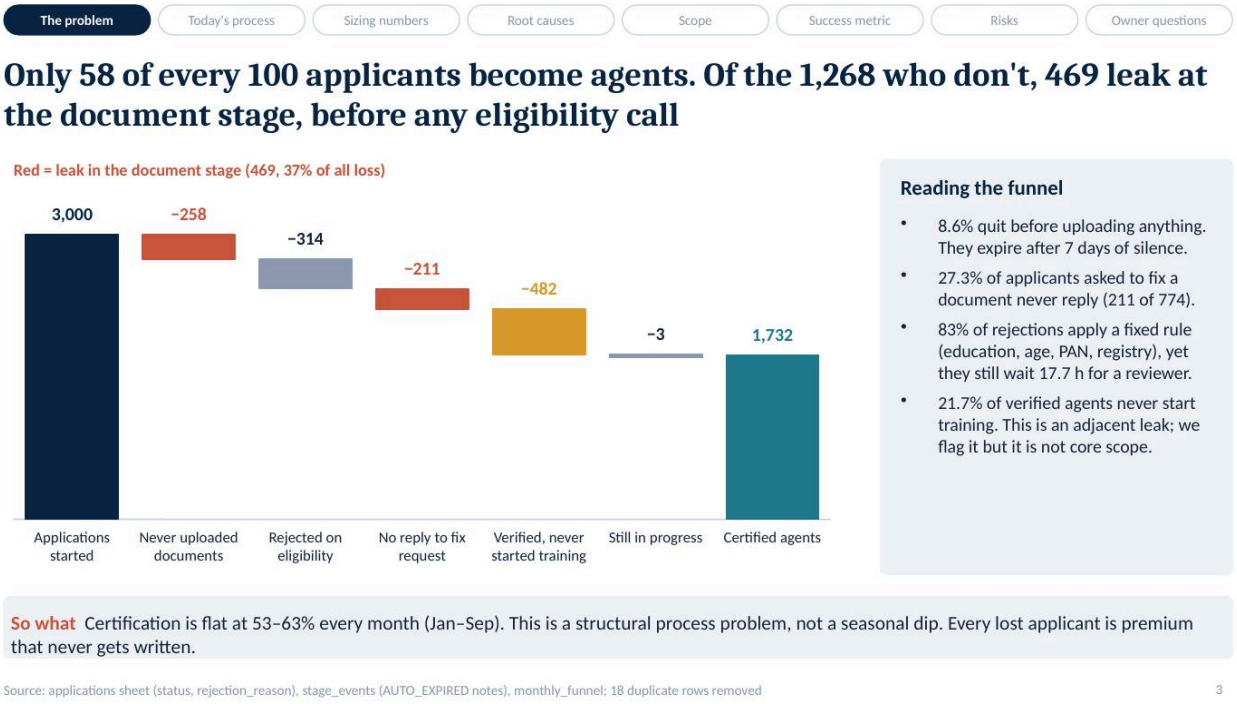


Say: "Three numbers: 57.7% certify, it takes 8.4 days, and files wait 17 hours before anyone looks, for a 20-minute check. Our five findings: where people leak, three root causes, and what AI should own, with a target of 60% decided straight-through without missing a single rejection."

IF A JUDGE ASKS "WHY LEAD WITH THE ANSWER?"

This is the pyramid principle: busy people want the conclusion first, then the evidence. Everything after this slide proves one line on this slide.

3 The problem: the funnel

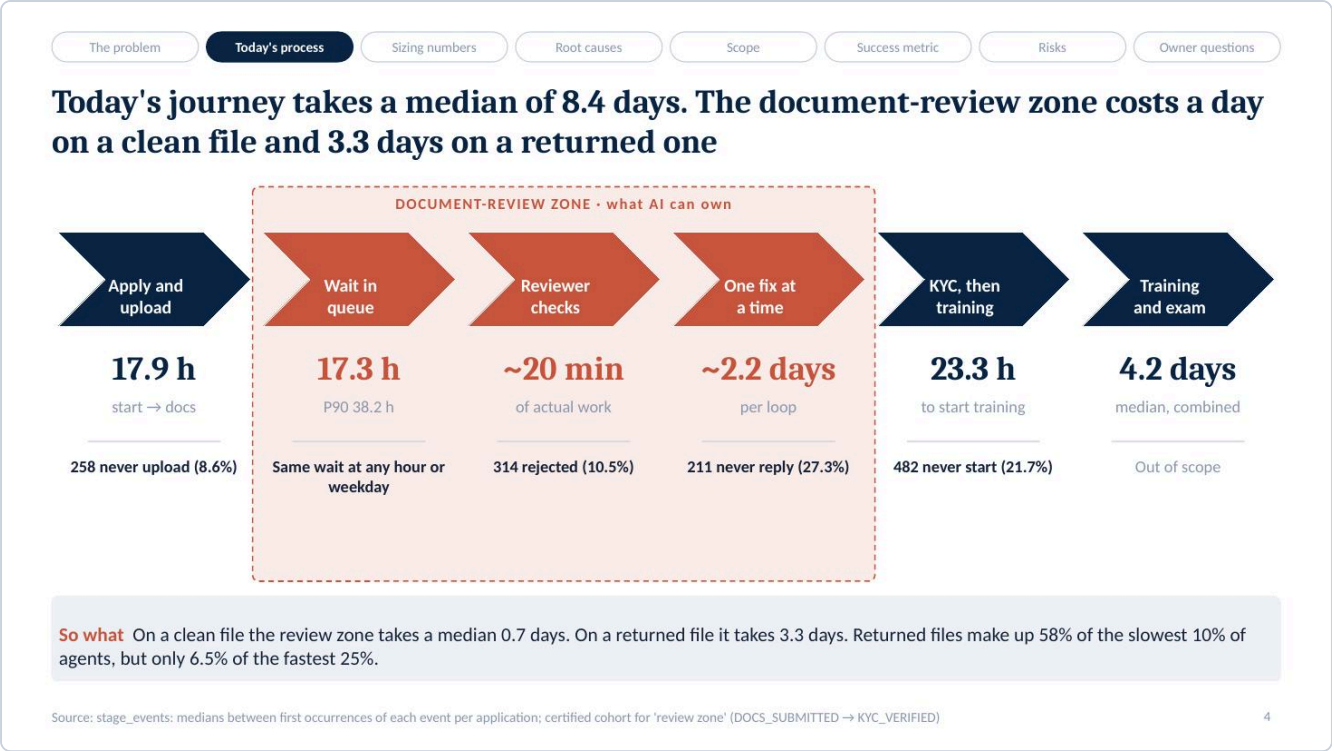


Say: "Start with 3,000. 258 never upload. 314 are rejected. 211 go silent after we ask for a fix. 482 pass KYC and never start training. That leaves 1,732. The red bars, 469 people, are lost in the document stage before anyone even decides whether they're eligible. And this isn't seasonal: every month sits between 53% and 63%."

IF A JUDGE ASKS "AREN'T REJECTIONS FINE? THOSE PEOPLE WEREN'T ELIGIBLE"

Yes, and that's why they're grey, not red. But 83% of them follow a fixed rule (age, education, PAN, registry) and still wait 17.7 hours to hear "no". A machine can tell them in minutes.

4 Today's process: the map



Say: "Here's the journey from the event log. The red zone is what AI can own: the 17-hour queue, a 20-minute check, and fix loops of about 2.2 days each. On a clean file this zone takes 0.7 days; on a returned file, 3.3. And 58% of the slowest agents had a return."

5 Today's process: where time goes



Say: "Compare the fastest quarter with the slowest tenth. The biggest gaps are the review zone (15 vs 76 hours) and training. Before the first decision, 98% of the time is waiting. And the wait is the same at 3 a.m. and 3 p.m., Monday or Sunday, so it's a backlog, not an off-hours gap. Training is out of scope, so we won't promise 2 days."

IF A JUDGE ASKS "SO YOUR TARGET ISN'T 2 DAYS LIKE THE BRIEF HINTS?"

Correct. Training alone is ~3 days and is out of scope. Removing the review-zone delay gets the median to ~6.9 days. Going further needs the training hand-off, which is owner question 5.

6 Sizing numbers



Say: "Eight numbers size the problem. The key point is the bottom box: reviewer labour on first review is only about ₹4 lakh for nine months. The prize isn't saving reviewer cost; it's the 469 people lost in the document stage and the days each bounce adds."

IF A JUDGE ASKS "HOW DID YOU GET ₹4.1 L?"

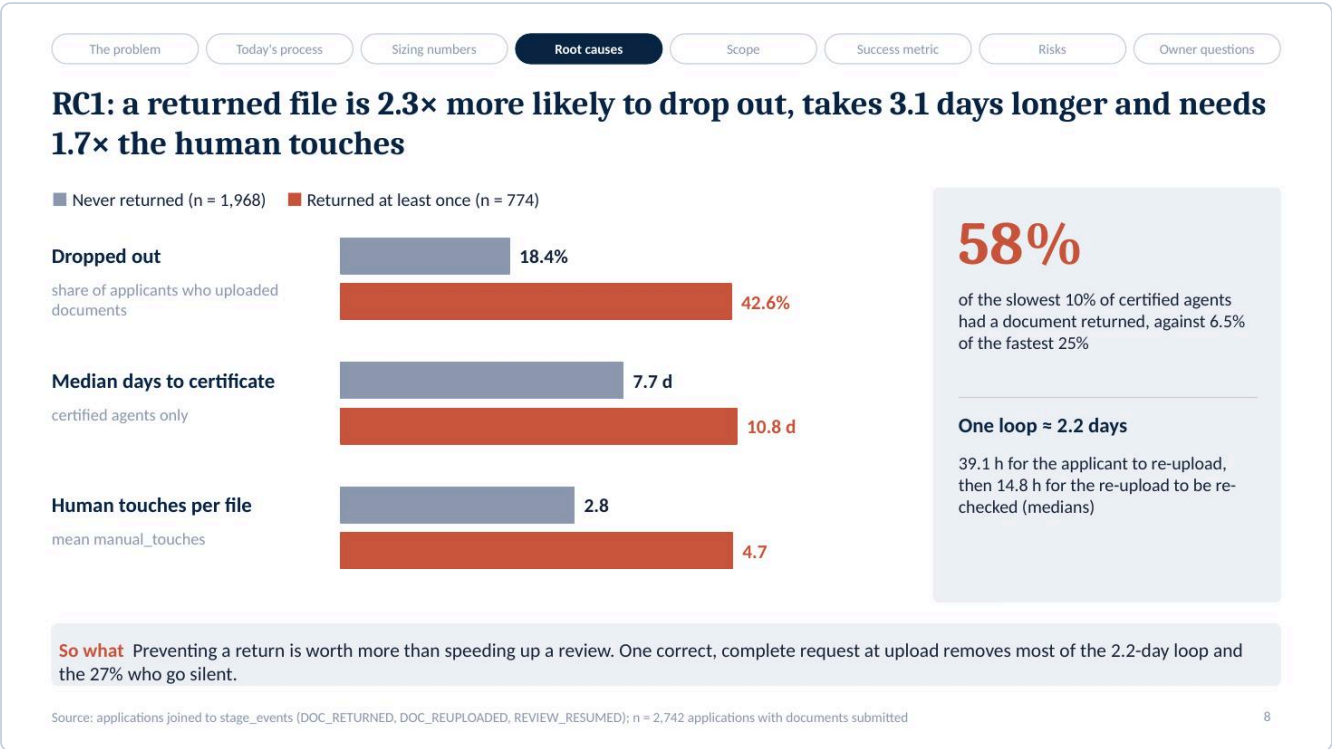
2,742 files × 20.7 minutes ≈ 946 hours × ₹430/hour (the reviewers' loaded cost weighted by files handled) ≈ ₹4.07 lakh. It covers first review only, not re-checks or other work.

7 Root causes: the issue tree



Say: "Why do 42% fail and why does it take 8.4 days? Four causes. Three sit inside the document check and AI can fix them. The fourth, people who never start training, we flag to the owner."

8 RC1: returned files

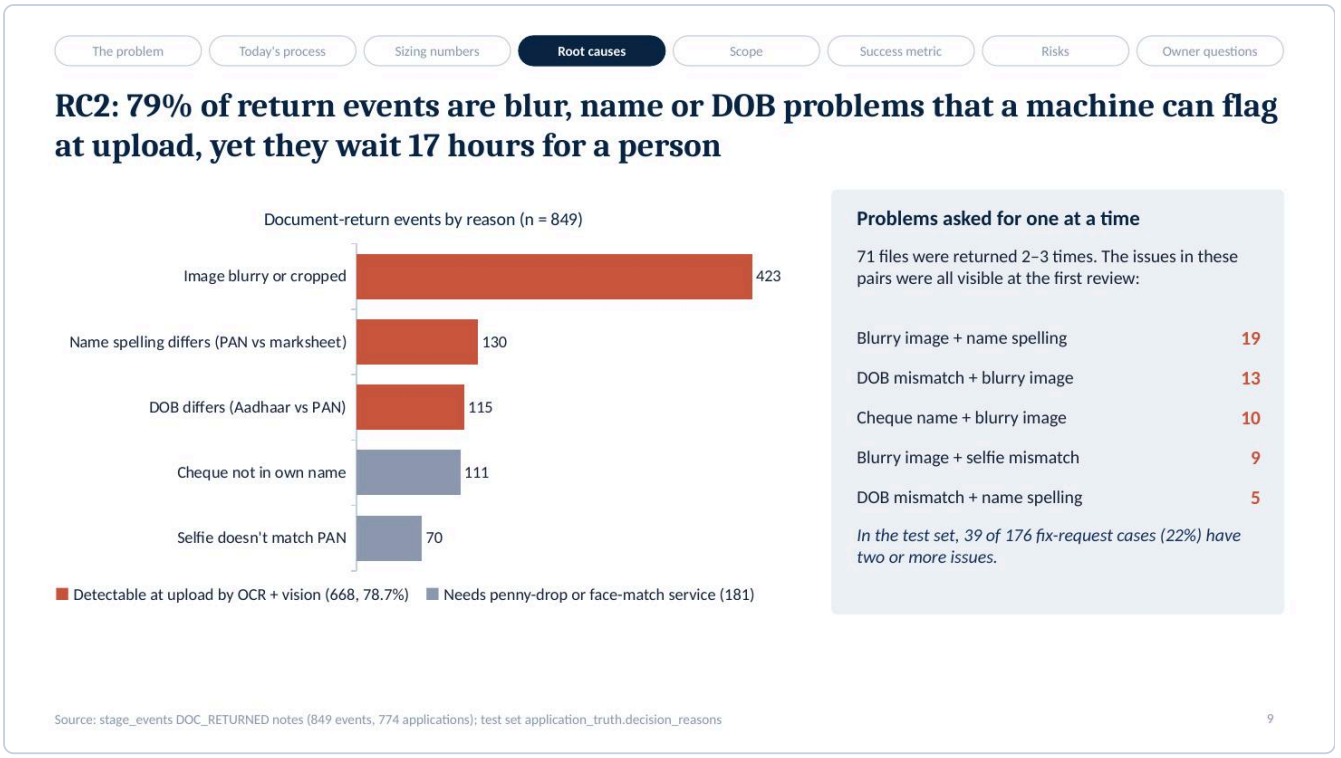


Say: "A returned file is 2.3 times more likely to drop out, 42.6% against 18.4%. It takes 3.1 days longer and needs 1.7 times the touches. One loop costs about 2.2 days. So preventing a return is worth more than speeding up a review."

IF A JUDGE ASKS "ISN'T THAT JUST CORRELATION? MAYBE WEAKER APPLICANTS GET RETURNED"

Fair. The data can't prove cause on its own. But the mechanism is direct: 211 people expired specifically because they didn't reply to a fix request. A pilot would confirm the size of the effect.

9 RC2: return reasons

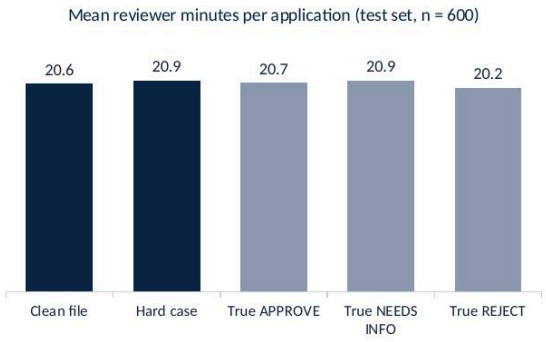


Say: "Of 849 return events, 79% are blur, name or DOB problems that can be flagged at upload. Half are just blurry photos. The 71 files returned more than once had problem pairs that were visible the first time, yet they were asked for one by one."

10 RC3: effort and names

The problem | Today's process | Sizing numbers | **Root causes** | Scope | Success metric | Risks | Owner questions

RC3: reviewers give every file the same ~21 minutes. Their judgement drifts on names, the one area that genuinely needs it



Effort is flat. 364 of 600 files (60.7%) are clean approvals that still get the full read.

Legitimate name forms that a simple string match would bounce

Name pattern vs PAN	Files	Correct decision: APPROVE
Initials (R K SHARMA)	28	18
Middle name missing	29	23
Order swapped	36	23
Middle name added	23	15
Total legitimate variants	**116**	**79**

- 79 of 364 good approvals (22%) carry a non-exact but legitimate name. Exact matching would wrongly stall 1 in 5 good agents.
- Reviewers coded 80 documents as "reviewer discretion" with no rule cited. 53 of those were on files the rulebook would approve.

Source: PS-01_document_review_test_set.xlsx: documents.reviewer_minutes, human_reason_code; application_truth.name_match_category, final_decision, is_hard_case

10

Say: "Reviewers give every file the same 21 minutes, clean or hard. Where judgement is genuinely needed, names, 22% of good approvals have a legitimate variant like initials or a swapped order. A naive string match would stall one in five good agents. And 53 approvable files were held under 'reviewer discretion' with no rule cited."

11 Scope

The problem | Today's process | Sizing numbers | Root causes | **Scope** | Success metric | Risks | Owner questions

Scope: AI owns the rule-based ~90% (approve, ask once, rule-based reject). People keep the calls the rulebook reserves for them

AI decides
end to end, with every rule cited

60.7% + 29.3%

of test-set files are APPROVE or NEEDS MORE INFO

- Quality gate at upload (blur, glare)
- Read name, DOB and ID from every document
- Cross-check names (rulebook variants) and DOB
- PAN type, age ≥ 18, Class 10 pass
- Use registry and penny-drop results
- One fix message, English or Hindi

AI recommends, human confirms
evidence laid out side by side

≈ 2%

of test files (12 different-person names) + Human-only rules

- REJECT for a different person's name (never auto-reject)
- Borderline face-match scores (e.g. 0.68)
- Any decision the rulebook marks Human only
- Cases where AI confidence is low

Out of scope
for this challenge

- Building the 15-hour training or the exam
- Real Aadhaar e-KYC, penny-drop or IRDAI calls (we use data-pack values)
- Fraud investigation beyond flagging
- Storing or showing full Aadhaar numbers
- Adjacent (v2): nudging the 482 who never start training

So what In production, 261 of 314 rejections (83%) apply a fixed rule: education, registry, age or PAN. AI can decide these in minutes. Only name-mismatch rejections need a human.

Source: test set application_truth.final_decision (364 / 176 / 60), name_match_category = DIFFERENT (12); applications.rejection_reason; PS-01 brief §3, §8

11

Say: "AI owns approvals, fix requests and rule-based rejections, about 90% of files. People keep the calls the rulebook reserves for them, like a different person's name or a borderline face match. Training, real government calls and fraud investigation are out."

12 Success metric



Say: "One north star: straight-through rate, from 0% to at least 60%, because 60.7% of files are true approvals. One guardrail: catch at least 98% of rejections. Each supporting metric maps to a root cause. In our simulation, that takes the median from 8.4 to 6.9 days and keeps about 105 more agents."

13 Risks

The problem | Today's process | Sizing numbers | Root causes | Scope | Success metric | **Risks** | Owner questions

The biggest risk is a confident wrong approval, and the data shows exactly where it would hide

Risk	Evidence from the data	Mitigation	Impact
Wrongful approval (compliance breach)	22 of 60 true rejections come from application-level rules (age, registry, PAN type). No single document shows them, so a document-by-document check would pass 9.	Run deterministic rule checks before any LLM call. Gate releases on REJECT recall ≥ 98%.	High
Wrongful rejection of a good agent	79 of 364 approvals (22%) carry legitimate name variants; Mohammad/Mohammed-type spellings are 36 more cases.	Encode the rulebook's name rules explicitly. When unsure, prefer NEEDS MORE INFO over REJECT.	High
Noisy photos and OCR	OCR confidence falls to 0.60 on blurry, glare or cropped images (0.93 when OK); light noise such as O/0 in ~25% of good files.	Quality gate at upload, per-field confidence, ask for re-capture before review.	Med
Uneven performance across groups	Certification today ranges from 53% (Odisha) to 65% (Delhi).	Report accuracy by state, gender and name pattern; publish any gap.	Med
Privacy and applicant trust	Aadhaar data under DPDP Act 2023; 27% already go silent after a fix request.	Mask Aadhaar before any model, log or prompt; plain-language messages; never mention fraud.	Med

So what Also, 2.5% of reviewer document verdicts are wrong. We will score the engine against ground truth, not against reviewers, or we will copy their mistakes.

Source: Test set: application_truth vs roll-up of document human_verdict; documents.ocr_confidence by image_quality; applications by state; PS-01 brief §8

13

Say: "The biggest risk is a confident wrong approval. 22 of 60 true rejections come from rules no single document shows, so they must be explicit checks, not left to an LLM. Second, wrongly rejecting good agents over name variants. And we grade against ground truth, not reviewers, because 2.5% of reviewer verdicts are wrong."

14 Owner questions

The problem | Today's process | Sizing numbers | Root causes | Scope | Success metric | Risks | **Owner questions**

Seven questions for the problem owner. Each one tests a hypothesis the data raised but cannot settle

	QUESTION	WHY WE ASK (EVIDENCE)
1	**What error rate on auto-approvals would compliance sign off, and who signs?**	Sets the straight-through ceiling. Reviewers run at 97.5% on documents today.
2	**Why does a 20-minute check wait 17 hours: staffing, batching or SLA design?**	The wait is identical at every hour and on every weekday.
3	**Are registry and penny-drop checks run before or after review, and what does each call cost?**	22 of 60 rejections depend on them; rejections take 17.7 h.
4	**What does "reviewer discretion" usually mean, and where do reviewers disagree most?**	80 discretion codes with no rule cited; 53 sit on approvable files.
5	**Is the hand-off to training inside our remit?**	482 verified agents (21.7%) never start training. It is the largest single leak.
6	**Can the pilot pull documents from DigiLocker instead of photos?**	Half of all returns (49.8%) are image-quality problems.
7	**Which region would you pilot in, and should applicants be told an AI decided?**	Certification varies 53–65% by state; 27% go silent after a request.

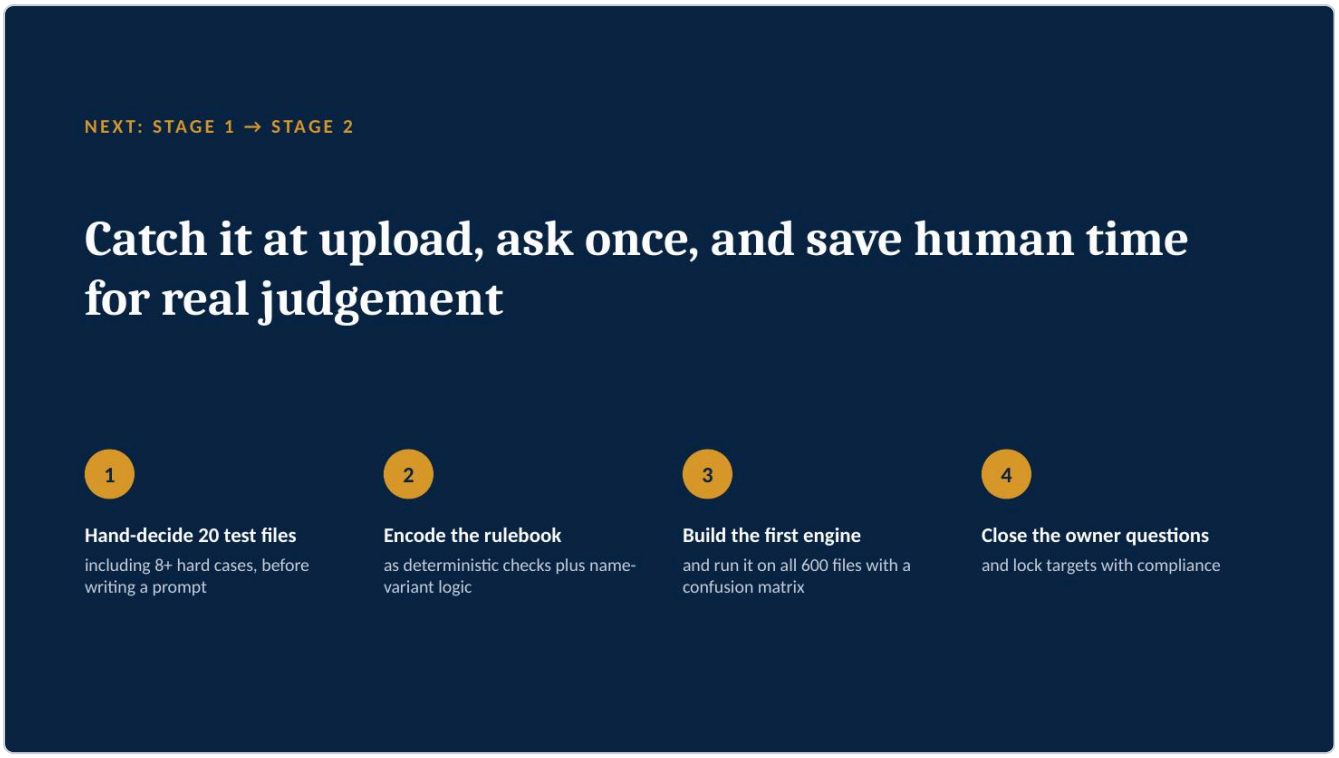
Red = must answer before we fix targets. They decide how much the engine may do on its own.

Source: Team analysis of PS-01 data pack; question list builds on PS-01 brief §10

14

Say: "Seven questions, each tied to something the data raised but can't settle. The first three, compliance tolerance, why the queue exists, and when registry checks run, decide how much our engine may do alone."

15–16 Next steps and appendix



Say: "Next: hand-decide 20 files including 8 hard ones, encode the rulebook, run a first engine on all 600, and lock targets with compliance." The appendix lists every definition, so any number can be traced.

PART 9

How a consultant builds a deck

Why our deck looks the way it does. Use these ideas for every future pitch.

Principle	What it means	Where you see it
Pyramid principle	Answer first, then the reasons, then the evidence. Busy people may stop reading after slide 2.	Executive summary is slide 2
Action titles	Every title is a full sentence that states the finding, not a topic label. Reading only the titles should tell the whole story.	"RC1: a returned file is 2.3× more likely to drop out..." instead of "Returns analysis"
One message per slide	If a slide needs two titles, it's two slides.	RC1, RC2 and RC3 each get their own slide
MECE	"Mutually exclusive, collectively exhaustive": no overlaps, no gaps.	The four root causes; the three scope columns
"So what"	Every chart ends with the implication, not just the fact.	Grey "So what" box at the bottom of each slide
Source on every slide	Every number is traceable. Credibility comes from being checkable.	Small grey source line at the bottom
Colour has meaning	Navy = neutral; red = problem / AI zone; teal = solution / target; amber = caution / adjacent. Never decorative.	Red bars = document-stage leak; teal = certified
Tracker	A small chip row so the audience always knows where they are.	The 8 chips at the top
Honesty about limits	Saying what you *can't* claim builds trust with judges.	"AI alone cannot reach 2 days"; simulation flagged as directional
Native, editable charts	Charts are real PowerPoint charts, so you can change them in seconds.	Slides 5, 9, 10

LIKE YOU'RE FIVE

A good deck is like a good story: tell the ending first ("the treasure is under the tree"), then show the clues one by one, and every clue points back to the tree.

Speaking tips

- Read the title, point at the evidence, say the "so what". About a minute per slide.
- Use one number per sentence. "Returned files drop out 2.3 times as often" beats reading three percentages.
- If you don't know, say "we didn't measure that; here's how we would" rather than guessing.

PART 10

Tough questions and answers

Question	Answer
Why not just hire more reviewers to kill the queue?	It would cut the 17-hour wait, but not the one-at-a-time loops (2.2 days each), the 27% who go silent, or the 21 minutes spent on clean files. More people doing the same process gives the same leaks.
Your data is synthetic. Why trust it?	It's designed to mirror the real problem, and the owner set it. We use it to find patterns and size them; a pilot would confirm the real sizes. We never claim it's real InsuranceDekho data.
Why 60% straight-through and not 80%?	60.7% of test files are true approvals. Going above that would mean auto-sending fix requests or rejections without review. That's possible for rule-based cases, but compliance must agree first (owner question 1).
Isn't an AI that rejects people dangerous?	Yes, which is why the AI never auto-rejects on name mismatch, every decision cites a rule, and "unsure" becomes NEEDS_MORE_INFO, not REJECT. The guardrail is 98% REJECT recall.
How is REJECT recall different from accuracy?	Accuracy is "how many answers are right overall". Recall on REJECT is "of the people who *should* be rejected, how many did we catch (or send to a human)". You can be 95% accurate and still miss a third of the rejections, because they're only 10% of files.
Why use an LLM at all? Why not just rules?	Rules handle age, PAN format, education and the registry perfectly, and we'd use them. But reading noisy OCR and judging "R K SHARMA = RAHUL KUMAR SHARMA" across messy text is where an LLM beats hand-written rules. Best of both: rules first, LLM for reading and fuzzy matching, human for judgement.
What about people without smartphones or with bad cameras?	The quality gate tells them instantly to retake. DigiLocker (stretch goal) removes photos entirely. Field partners already bring in 31% of applicants and can help.
How will you handle Hindi?	The single follow-up message is generated in English or Hindi, as the brief requires. We'll test clarity with five users (target ≥ 4/5).
What if the model is wrong and nobody notices?	Audit trail for every decision, a reviewer dashboard showing where the AI was unsure, compliance sampling, and fairness reports by state, gender and name pattern.
Why does the queue wait not depend on the time of day?	We don't know yet; it suggests a standing backlog or batch processing rather than missing night shifts. That's owner question 2.
Where did "2.3×" come from?	42.6% (330 of 774 returned) ÷ 18.4% (363 of 1,968 not returned), among the 2,742 who uploaded documents.

Is the 6.9-day figure a promise?	No. It's a simulation: we replaced each certified agent's review-zone time with 10 minutes and kept everything else. It's directional and needs a pilot.
What's the cost of the AI per application?	The brief says "a few rupees per application" for current models. External calls (penny-drop, registry) are extra; owner question 3 asks their cost.
Why only 7 owner questions when the brief lists 9?	We kept the ones our data sharpened and attached evidence to each. The brief's others (whose onboarding costs most; reviewer's day) are good warm-ups if time allows.

PART 11

Glossary A to Z

Every term, in words a five-year-old would get, plus the grown-up meaning.

A

Aadhaar
India's 12-digit ID card number. *Simple:* your "who am I and where do I live" card. We may only ever see the last 4 digits.

Accuracy
Share of all decisions that are correct. *Simple:* how many answers you got right on the test.

Action title
A slide title written as a full-sentence finding. *Simple:* the title tells you the answer, not just the topic.

Agent / applicant
The person trying to become a POSP.

Application-level rule
A rule you can only check by looking at the whole application, not one document (e.g. registry hit, age on application date).

Audit trail
A record of every decision, the rule behind it and the model version. *Simple:* a diary that shows why the robot said yes or no.

AUTO_EXPIRED
The system closes a file after a deadline passes (7, 10 or 14 days of no action).

B

Backlog
A pile of work waiting to be done. *Simple:* the queue at the ice-cream shop.

Baseline
Today's number, before any change. You need it to show improvement.

BANK_NAME_MISMATCH
The cheque's account isn't in the applicant's own name (e.g. a parent's).

C

Cancelled cheque
A cheque with "cancelled" written on it, used to prove bank details.

Certification rate
Certified ÷ started. Ours: 57.7%.

Compliance
Following the regulator's rules. A wrongful approval is a "compliance breach".

Confidence (field-level)
How sure the robot is about each thing it read (0 to 1).

Confusion matrix
A table of "what the right answer was" vs "what we said". *Simple:* a scoreboard of which answers got mixed up with which.

Consolidated message
One message listing every fix needed. *Simple:* tell them everything at once.

Cross-document check
Comparing the same field (name, DOB) across different documents.

D

DigiLocker
The government's online locker of verified documents. *Simple:* a safe cupboard where your papers already live, so you don't need to photograph them.

DOB
Date of birth.

DPDP Act 2023
India's Digital Personal Data Protection law. *Simple:* "only take what you need and keep it safe".

Drop-off / drop-out
People who leave before finishing.

Deduplicate
Remove repeated rows so nobody is counted twice.

E – F

Event log
A list of everything that happened, each with a time stamp (our stage_events sheet).

Executive summary
The one slide that says the whole answer.

Face-match score
0–1 similarity between the selfie and the PAN photo. Below 0.60 = FACE_MATCH_LOW; borderline scores (e.g. 0.68) go to a human.

False approval / false rejection
Saying yes when it should be no / no when it should be yes.

Funnel
How many people remain at each step. *Simple:* a funnel is wide at the top and narrow at the bottom, like people dropping out along the way.

G – H

Ground truth
The correct answer (application_truth.final_decision). *Simple:* the answer key at the back of the book.

Guardrail
A limit you must never cross while chasing the main goal.

Hard case
A file where a careful reviewer has to think (name variants, DOB conflicts, state differences). 199 of 600.

Human-in-the-loop
A design where people review or approve some of the system's decisions.

Human only
Decisions the rulebook reserves for a person.

Human touch / manual_touches
Number of times a person acted on a file. Average 3.3.

I – K

IRDAI
Insurance Regulatory and Development Authority of India.

IRDAI registry hit
The PAN is already registered as a POSP elsewhere, so the applicant must be rejected (ALREADY_REGISTERED).

Issue tree
A diagram that breaks one big "why?" into smaller causes.

KYC
Know Your Customer: checking someone is who they say they are.

L – M

LLM
Large language model, the kind of AI that reads and writes text.

Loaded cost
Salary plus overheads per hour (₹380–₹480; we use a weighted ₹430).

Mean
The average: add everything up and divide by how many.

Median
The middle value when everything is lined up. *Simple:* line 11 kids up by height; the 6th kid is the median.

MECE
Mutually exclusive, collectively exhaustive: no overlaps, no gaps.

Multimodal model
An AI that understands pictures and text together.

N – O

NAME_MISMATCH
A different person's name on a document. Recommend REJECT; a human must confirm.

NAME_SPELLING_VARIANT
Same person, spelt differently (MOHAMMED / MOHAMMAD). Ask for a corrected document or affidavit.

NEEDS_MORE_INFO
Fixable problem; ask the applicant.

North star metric
The one number the whole team steers by. Ours: straight-through rate.

OCR
Optical character recognition: turning a picture of text into text. *Simple:* the robot's eyes for reading.

Onboarding
Everything from "I want to join" to "you're certified".

P

P90 / P95 (percentile)
90% (or 95%) of cases are at or below this value. *Simple:* "almost everyone" is faster than this; it shows the slow tail.

PAN
Permanent Account Number: 10-character tax ID in the pattern AAAAA9999A. The 4th character P means an individual; C means a company.

Penny-drop
Sending ₹1 to a bank account to check it exists and get the holder's name.

Pilot
A small real-world trial (e.g. one region) before full launch.

POSP
Point of Sales Person: a part-time insurance agent.

Precision
Of the files we *said* were REJECT, how many truly were. *Simple:* when you shout "wolf!", how often there really is a wolf.

Pyramid principle
Answer first, then supporting reasons, then evidence.

Q – R

Queue wait
Time from upload to a reviewer opening the file. Median 17.3 h.

Recall
Of the files that truly *were* REJECT, how many we caught. *Simple:* of all the wolves out there, how many you spotted.

Reason code
A short code for why a document was flagged (14 of them).

Return / DOC_RETURNED
A reviewer sends a document back for a fix.

Review zone
Our term: upload → KYC verified. The part AI can own.

REVIEWER_DISCRETION
A reviewer's judgement with no standard rule code.

Root cause
The deepest "why" that, if fixed, stops the problem.

Rulebook
PolicyNest's 10-section onboarding standard; it defines a correct decision.

S – Z

Simulation
A "what if" calculation on today's data. Directional, not a promise.

SLA (service level)
A promised time limit, e.g. "first review within 2 hours".

So what
The implication of a fact: why the audience should care.

Specimen documents
12 worked example cases in the pack (not in our upload).

Straight-through processing (STP)

A file handled start to finish with no human touch. *Simple:* the robot does it all, correctly.

Synthetic data

Realistic but made-up data that protects privacy.

Test set

The 600 labelled applications we grade our engine on.

Waterfall chart

A chart showing a starting number shrinking step by step (our funnel slide).

PART 12

Caveats and unknowns

Say these before a judge does. It makes everything else more believable.

1. **Synthetic data.** Patterns are meaningful, but exact sizes need a pilot.
2. **Correlation is not causation.** Returned files drop out more, but some applicants may be both more likely to be returned and more likely to quit. The 211 "no reply" expiries are the direct link.
3. **Event-level vs person-level.** 849 return *events* come from 774 *people*. Percentages like 79% are event-level.
4. **Return events ≠ WhatsApp messages.** We assume each return generated a message; message logs would confirm it.
5. **Reviewer cost is first review only.** ₹4.1 L excludes re-checks and other tasks, so the 22-person team does more than this.
6. **Reviewer roll-up is our construction.** We combined document verdicts into one application verdict to show what a document-only check misses. Real reviewers may check age and the registry separately.
7. **The rulebook wasn't in our upload.** Exact face-match bands and the "Human only" list must be read from it before Stage 2.
8. **Simulation is directional.** It assumes saved time flows straight through, and that 78.9% of repliers still certify.
9. **Event times are around the clock.** Review pick-ups are logged at every hour despite day shifts. That's either a data artefact or a sign of batch logging; ask the owner.

PART 13

What comes next: Stage 2 in plain words

LIKE YOU'RE FIVE

We build the robot helper in four layers, like a sandwich: **eyes** (read the papers), **rule book** (check the must-be-true things), **brain** (compare tricky names and dates), and a **bell** that rings for a grown-up when the robot isn't sure.

#	Layer	What it does	Proves which test
1	Quality gate	Rejects blurry, cropped or glare photos at upload; asks for a retake immediately	T5; RC2
2	Extraction	Reads name, DOB, ID from OCR text (LLM tolerant of O/0 noise), with confidence per field	T1
3	Deterministic rules	PAN pattern and 4th char, age ≥ 18, Class 10 pass, registry hit, penny-drop name, face score thresholds	T6, T8; risk 1
4	Cross-document reasoning	Name variants per rulebook (initials, swap, middle), DOB agreement; state difference = informational	T2, T3, T7, T9
5	Decision and message	APPROVE / NEEDS_MORE_INFO / REJECT with every rule cited; one consolidated message in English or Hindi	T5
6	Human routing	NAME_MISMATCH and Human-only cases go to the reviewer view with evidence side by side	T4, T8
7	Re-check and audit	Re-validate only the changed document; log both rounds	T10

Stage 1 checklist (from the brief)

- Process map from stage_events, with time and drop-off at each step. **Done** (slides 4–5).
- 20 test applications decided by hand, including at least 8 hard cases. **Next.**
- The decisions we automate and the ones we route to a human. **Done** (slide 11).
- First business case: hours and drop-offs saved. **Done** (slides 6, 12).
- Answers from the problem-owner meeting. **Questions ready** (slide 14).

APPENDIX

The analysis code

One script makes every number in this book and the deck. Run it whenever you need to prove or update a number.

How to run

```
pip install pandas openpyxl
python ps01_analysis.py PS-01_posp_applications.xlsx PS-01_document_review_test_set.xlsx
```

What each block does, and why

Block	What	Why it's done this way
0	Load and de-duplicate	The README warns of 18 duplicate rows. Counting them twice would inflate every rate.
—	The "first" table	Turns the long event log into one row per person and one column per event (the first time it happened). Then any time gap is just column B − column A. This is the core trick of process mining.
1	Funnel	Counts final statuses and reads the AUTO_EXPIRED notes to see *why* people left.
2	Step times	Medians (typical person) and P90 (slow tail) per step. Compares fastest 25% vs slowest 10% to find where extra time goes. Checks wait by hour and weekday to test the "nobody works at night" idea.
3	Correction loop	Splits people into "ever returned" vs "never returned" and compares drop-out, days and touches. This isolates the cost of a return.
4	Sizing	Weighted reviewer cost (busier reviewers count more), touches, state and monthly spread.
5	Test set	Decision mix, name patterns, reviewer minutes by difficulty, multi-issue cases, OCR confidence, and the roll-up of reviewer verdicts against ground truth.
6	Simulation	Replaces each certified person's review-zone time with 10 minutes and recomputes the median, then estimates extra agents if silence fell to 10%.

The full script

```
"""
PS-01 · AI-First POSP Onboarding — the one script behind every number in the deck.

HOW TO RUN
    pip install pandas openpyxl
    python ps01_analysis.py  <path to PS-01_posp_applications.xlsx>  <path to PS-
01_document_review_test_set.xlsx>
```

```
WHY ONE SCRIPT
    If a judge asks "where does 42.6% come from?", you open this file, find the
    numbered block, and show the exact rows and the exact formula.
"""
import sys
import pandas as pd

APPS_FILE, TEST_FILE = sys.argv[1], sys.argv[2]

# ---------------------------------------------------------------------------
# 0. LOAD + CLEAN
#    The README says 18 applications appear twice. We keep the first copy of
#    each application_id so nobody is counted twice (3,018 -> 3,000 rows).
# ---------------------------------------------------------------------------
X = pd.read_excel(APPS_FILE, sheet_name=None)
apps = X["applications"].drop_duplicates("application_id").copy()
ev = X["stage_events"].copy()
ops = X["ops_team"]
ev["t"] = pd.to_datetime(ev["event_time"])
print("Applications after de-duplication:", len(apps))

# "first" = a table with one row per application and one column per event,
# holding the FIRST time that event happened. Every time gap below is just
# (later column - earlier column).
first = ev.sort_values("t").groupby(["application_id", "event"])["t"].first().unstack()


def hours(df, a, b):
    """Hours between event a and event b for each application."""
    return (df[b] - df[a]).dt.total_seconds() / 3600


# ---------------------------------------------------------------------------
# 1. THE FUNNEL (slide 3)
# ---------------------------------------------------------------------------
print("\n[1] Final status:\n", apps["status"].value_counts())
expired = ev[ev.event == "AUTO_EXPIRED"]["note"].value_counts()
print("\n[1] Why people were auto-expired:\n", expired)
print("\n[1] Rejection reasons:\n", apps["rejection_reason"].value_counts())

# ---------------------------------------------------------------------------
# 2. TIME BETWEEN STEPS (slides 4-5)
# ---------------------------------------------------------------------------
steps = [("APPLICATION_STARTED", "DOCS_SUBMITTED"), ("DOCS_SUBMITTED", "REVIEW_STARTED"),
         ("REVIEW_STARTED", "KYC_VERIFIED"), ("KYC_VERIFIED", "TRAINING_STARTED"),
         ("TRAINING_STARTED", "TRAINING_COMPLETED"), ("TRAINING_COMPLETED", "EXAM_PASSED"),
         ("EXAM_PASSED", "CERTIFIED"), ("DOC_RETURNED", "DOC_REUPLOADED"),
         ("DOC_REUPLOADED", "REVIEW_RESUMED")]
print("\n[2] Median / P90 hours per step (all applications that reached both events)")
for a, b in steps:
    h = hours(first, a, b).dropna()
    print(f"   {a:>20} -> {b:<20} median {h.median():6.1f}h  P90 {h.quantile(.9):6.1f}h  n={len(h)}")

cert = first[first.CERTIFIED.notna()].copy()
total_h = hours(cert, "APPLICATION_STARTED", "CERTIFIED")
print(f"\n[2] Days to certificate: median {total_h.median()/24:.1f}, "
      f"P25 {total_h.quantile(.25)/24:.1f}, P90 {total_h.quantile(.9)/24:.1f}")

# Fast quartile vs slow decile: where does the extra time go?
fast = cert[total_h <= total_h.quantile(.25)]
slow = cert[total_h >= total_h.quantile(.90)]
print("\n[2] Median hours: fastest 25% vs slowest 10%")
for a, b in steps[:7]:
    print(f"   {a}->{b}: fast {hours(fast, a, b).median():.1f}h  slow {hours(slow, a,
b).median():.1f}h")

# Is the queue wait worse at night or at weekends? (It isn't.)
w = first.copy()
w["wait"] = hours(w, "DOCS_SUBMITTED", "REVIEW_STARTED")
```

```
w["hour"] = w.DOCS_SUBMITTED.dt.hour
w["weekday"] = w.DOCS_SUBMITTED.dt.dayofweek
w["band"] = pd.cut(w.hour, [-1, 8, 17, 21, 23], labels=["00-09", "09-18", "18-22", "22-24"])
print("\n[2] Median queue wait by submission time band:\n", w.groupby("band",
observed=True).wait.median().round(1))
print("[2] Median queue wait by weekday (0=Mon):\n", w.groupby("weekday").wait.median().round(1))
print(f"[2] Queue wait P95: {w.wait.quantile(.95):.1f}h")

# ---------------------------------------------------------------------------
# 3. THE CORRECTION LOOP (slides 8-9)
# ---------------------------------------------------------------------------
returned_ids = set(ev[ev.event == "DOC_RETURNED"].application_id)
apps["returned"] = apps.application_id.isin(returned_ids)
uploaded = apps[apps.docs_submitted_at.notna()]
print("\n[3] Unique returned applications:", len(returned_ids), "| return events:", (ev.event ==
"DOC_RETURNED").sum())
print("[3] Drop-out rate, returned vs not (of applicants who uploaded):\n",
      uploaded.groupby("returned").status.apply(lambda s: (s == "DROPPED").mean()).round(3))
print("[3] Median days to certificate, returned vs not:\n",
(apps.groupby("returned").hours_to_certify.median() / 24).round(1))
print("[3] Mean human touches, returned vs not:\n",
apps.groupby("returned").manual_touches.mean().round(2))
print("[3] Share of touches from returned files:",
      round(apps[apps.returned].manual_touches.sum() / apps.manual_touches.sum(), 3))
cert["returned"] = cert.index.isin(returned_ids)
print(f"[3] Returned share: slowest 10% {cert.loc[slow.index,'returned'].mean():.3f}, "
      f"fastest 25% {cert.loc[fast.index,'returned'].mean():.3f}")
print("[3] Review zone (docs -> KYC) median days, returned vs not:\n",
      (hours(cert, "DOCS_SUBMITTED", "KYC_VERIFIED").groupby(cert.returned).median() / 24).round(2))
print("\n[3] Return reasons (event level):\n", ev[ev.event == "DOC_RETURNED"].note.value_counts())
per_app = ev[ev.event == "DOC_RETURNED"].groupby("application_id").size()
print("[3] Returns per application:\n", per_app.value_counts())

# ---------------------------------------------------------------------------
# 4. SIZING (slide 6)
# ---------------------------------------------------------------------------
cost = (ops.loaded_cost_inr_per_hour * ops.applications_handled_jan_sep).sum() /
ops.applications_handled_jan_sep.sum()
print(f"\n[4] Workload-weighted reviewer cost: Rs {cost:.0f}/h; touches total
{apps.manual_touches.sum()}, "
      f"mean {apps.manual_touches.mean():.2f}")
print("[4] Certification rate by state:\n", apps.groupby("state").status.apply(lambda s: (s ==
"CERTIFIED").mean()).round(3).sort_values())
print("[4] Monthly funnel:\n", X["monthly_funnel"][["month", "applications", "certification_rate"]])

# ---------------------------------------------------------------------------
# 5. THE LABELLED TEST SET (slides 10, 11, 13)
# ---------------------------------------------------------------------------
T = pd.read_excel(TEST_FILE, sheet_name=None)
docs, truth = T["documents"], T["application_truth"]
print("\n[5] Correct decisions:\n", truth.final_decision.value_counts())
print("[5] Name pattern x decision:\n", pd.crosstab(truth.name_match_category, truth.final_decision))
mins = docs.groupby("application_id").reviewer_minutes.sum().rename("mins")
tm = truth.merge(mins, left_on="application_id", right_index=True)
print(f"[5] Reviewer minutes per application: mean {tm.mins.mean():.1f}")
print("[5] ...by hard case:\n", tm.groupby("is_hard_case").mins.mean().round(1))
print("[5] ...by correct decision:\n", tm.groupby("final_decision").mins.mean().round(1))
n_reasons = truth.decision_reasons.fillna("").str.count(";") +
truth.decision_reasons.notna().astype(int)
print("[5] NEEDS_MORE_INFO cases with 2+ reasons:", ((truth.final_decision == "NEEDS_MORE_INFO") &
(n_reasons >= 2)).sum())
print("[5] OCR confidence by image quality:\n",
docs.groupby("image_quality").ocr_confidence.mean().round(2))

# Roll the reviewers' document verdicts up to one application verdict:
# any REJECT -> REJECT; else any REUPLOAD/NEEDS_INFO -> NEEDS_MORE_INFO; else APPROVE.
rank = {"ACCEPT": 0, "REUPLOAD": 1, "NEEDS_INFO": 1, "REJECT": 2}
rolled = (docs.assign(r=docs.human_verdict.map(rank)).groupby("application_id").r.max()
          .map({0: "APPROVE", 1: "NEEDS_MORE_INFO", 2: "REJECT"}))
```

```
tm["rolled"] = tm.application_id.map(rolled)
print("[5] Ground truth (rows) vs rolled-up reviewer verdict (cols):\n", pd.crosstab(tm.final_decision,
tm.rolled))
disc = docs[docs.human_reason_code == "REVIEWER_DISCRETION"]
print("[5] REVIEWER_DISCRETION codes:", len(disc), "| on files truly APPROVE:",
      truth[truth.application_id.isin(disc.application_id)].final_decision.eq("APPROVE").sum())

# ---------------------------------------------------------------------------
# 6. SIMULATION (slide 12) — directional, not a forecast
# ---------------------------------------------------------------------------
sim = total_h - hours(cert, "DOCS_SUBMITTED", "KYC_VERIFIED") + 10 / 60   # review zone -> 10 minutes
print(f"\n[6] Median days to certificate: today {total_h.median()/24:.1f} -> simulated
{sim.median()/24:.1f}; "
      f"P90 {total_h.quantile(.9)/24:.1f} -> {sim.quantile(.9)/24:.1f}")
silent = set(ev[(ev.event == "AUTO_EXPIRED") & ev.note.str.contains("No response")].application_id)
replied = apps[apps.returned & ~apps.application_id.isin(silent)]
conv = (replied.status == "CERTIFIED").mean()
extra = (len(silent) - round(0.10 * len(returned_ids))) * conv
print(f"[6] Repliers who certify: {conv:.3f}; extra agents if silence falls to 10%: {extra:.0f}")
```